Day Trader Metrics

2026/06/18

Environment Setup

No API keys needed in this, no trades are actioned, only an analysis of public market data.

1. Data sourced live using `CCXT` API library, no scraping required.
2. Indicators derived using `pandas-ta-classic`, replacing `numpy.2`

Run the following cell each time a new kernel starts, or whenever you start a new session. To avoid setup issues across different users, the `venv` environment was removed but package list requirements.txt list was maintained. This cell builds everything again and configures the notebook graphics.

Make sure to log into Binance and Alpaca accounts before running this setup below. To avoid api issues, the prferred Binance.US login is accessed here: https://docs.ccxt.com/en/latest/exchange-markets.html

```
import sys, subprocess
from pathlib import Path

# assign Python kernel,
REQUIRED_PY = (3, 11)
if sys.version_info[:2] != REQUIRED_PY: raise RuntimeError(
        f"This notebook needs Python {REQUIRED_PY[0]}.{REQUIRED_PY[1]}, "
        f"but the kernel is {sys.version.split()[0]}. Switch the kernel and re-run.")

# Install packages & versions in requirements.txt
REQUIREMENTS = Path("inputs/requirements.txt")
if not REQUIREMENTS.exists():
    raise FileNotFoundError(f"{REQUIREMENTS} not found — run from the repo root.")
subprocess.run([sys.executable, "-m", "pip", "install", "--quiet",
     "--break-system-packages", "--disable-pip-version-check",
     "-r", str(REQUIREMENTS)], check=True,)

# import package modules and naming
```

```python
import warnings, numpy as np, pandas as pd
from datetime import datetime
import ccxt
import pandas_ta_classic as ta
from pykalman import KalmanFilter
import plotly, plotly.graph_objects as go, plotly.io as pio
warnings.filterwarnings("ignore")
pio.renderers.default = "plotly_mimetype+notebook_connected"

# quick sanity check
print(f"Environment ready  ·  python {sys.version.split()[0]}  ({sys.executable})")
print(f"  ccxt {ccxt.__version__} · pandas {pd.__version__} · "
      f"numpy {np.__version__} · plotly {plotly.__version__}")
```

```
Environment ready  ·  python 3.11.13  (/Users/seamus/.local/share/uv/python/cpython-3.11.13-macos-aarch64-none/bin/python)
  ccxt 4.5.59 · pandas 2.3.3 · numpy 2.4.6 · plotly 6.8.0
```

Import Data

Live price data is sourced between 24 hours and 1 week from multiple trading platforms. Five filters were highlighted below, including EXCHANGE, SANDBOX, TIMEFRAME, LIMIT, and TOP_N. Currently, theccxt library of APIs maintains access to 105 leading crypto exchange platforms. Happy browsing!

```python
EXCHANGE  = "binance"   # 105 platforms available, see https://github.com/ccxt/ccxt/wiki/Exchange-Markets)
SANDBOX   = False       # True = testnet / fake money (mainly for the trading step later)
TIMEFRAME = "1d"        # candle size: "1m","5m","1h","4h","1d","1w"
LIMIT     = 300         # candles pulled per coin (max 1000)
TOP_N     = 20          # how many of the most-traded /USDT pairs to scan

exchange = getattr(ccxt, EXCHANGE)()
exchange.set_sandbox_mode(SANDBOX)
print(f"Configured: {EXCHANGE} | sandbox={SANDBOX} | {TIMEFRAME} candles x{LIMIT} | top {TOP_N}")
```

```
Configured: binance | sandbox=False | 1d candles x300 | top 20
```

```python
def calculate_indicator(symbol, timeframe=TIMEFRAME, limit=LIMIT):
    bars = exchange.fetch_ohlcv(symbol, timeframe=timeframe, limit=limit)
    df = pd.DataFrame(bars[:-1], columns=["timestamp","open","high","low","close","volume"])
    df["timestamp"] = pd.to_datetime(df["timestamp"], unit="ms")
    df = df.set_index("timestamp")
    close, low = df["close"].iloc[-1], df["low"].iloc[-1]

    # Kalman threshol & smoother
    kf = KalmanFilter(transition_matrices=[1], observation_matrices=[1],
                      initial_state_mean=0, initial_state_covariance=1,
                      observation_covariance=1, transition_covariance=.01)
    state_means, _ = kf.filter(df["close"].values)
    df["kf_mean"] = state_means
    kalman = df["kf_mean"].iloc[-1]
    above_kalman = bool(low > kalman)

    # Trend: Is EMA-14 leading the Kalman mean?
    df.ta.ema(length=14, append=True)
    ema_cross = bool(df["EMA_14"].iloc[-1] > kalman)

    # Bollinger(14) mean-reversion envelope.
    bb = df.ta.bbands(length=14)
    bbl, bbu = bb["BBL_14_2.0"].iloc[-1], bb["BBU_14_2.0"].iloc[-1]

    # Ichimoku projectionse.
    ich = df.ta.ichimoku()[1]
    isa_9, isb_26 = ich["ISA_9"].iloc[-1], ich["ISB_26"].iloc[-1]

    # Archer MA trend flag, RSI, Choppiness.
```

```
    amat = bool(df.ta.amat()["AMATe_LR_8_21_2"].iloc[-1] == 1)
    rsi = float(df.ta.rsi().iloc[-1])
    chop = round(float(df.ta.chop().iloc[-1]), 2)

    # Candle shape check: any doji / dragonfly / gravestone candles?
    o, h, l, c = df["open"].iloc[-1], df["high"].iloc[-1], df["low"].iloc[-1], df["close"].iloc[-1]
    rng   = h - l
    body  = abs(c - o)
    upper = h - max(o, c)
    lower = min(o, c) - l
    doji       = bool(rng > 0 and body <= 0.10 * rng)
    dragonfly  = bool(doji and lower >= 0.6 * rng and upper <= 0.10 * rng)
    gravestone = bool(doji and upper >= 0.6 * rng and lower <= 0.10 * rng)

    # MACD (12,26,9)
    macd_line   = df["close"].ewm(span=12, adjust=False).mean() - df["close"].ewm(span=26,
adjust=False).mean()
    signal_line = macd_line.ewm(span=9, adjust=False).mean()
    macd, sig           = macd_line.iloc[-1], signal_line.iloc[-1]
    macd_prev, sig_prev = macd_line.iloc[-2], signal_line.iloc[-2]
    macd_buy  = bool(macd_prev <= sig_prev and macd > sig)
    macd_sell = bool(macd_prev >= sig_prev and macd < sig)
    macd_below_zero = bool(macd < 0)
    buy  = amat and ema_cross and above_kalman
    sell = (not amat) and (not ema_cross) and (not above_kalman)
    row = dict(Symbol=symbol, Buy=buy, Sell=sell, Close=round(float(close), 4),
               RSI=round(rsi, 2), Chop=chop, AMAT=amat,
               Ichimoku_9=round(float(isa_9), 4), Ichimoku_26=round(float(isb_26), 4),
               EMA_gt_Kalman=ema_cross, Low_gt_Kalman=above_kalman,
               Doji=doji, Dragonfly=dragonfly, Gravestone=gravestone,
               MACD=round(float(macd),4), MACD_Signal=round(float(sig),4),
               MACD_Buy=macd_buy, MACD_Sell=macd_sell, MACD_below_zero=macd_below_zero)

    return df, row

def plot(symbol):
    df, _ = calculate_indicator(symbol)
    fig = go.Figure(go.Candlestick(x=df.index, open=df.open,
        high=df.high, low=df.low, close=df.close, name=symbol))
    fig.add_trace(go.Scatter(x=df.index, y=df["kf_mean"],
         name="Kalman", line=dict(color="orange", width=2), opacity=0.7))
    fig.add_trace(go.Scatter(x=df.index, y=df["EMA_14"],
        name="EMA-14", line=dict(color="purple", width=2), opacity=0.7))
    fig.update_layout(title=symbol, xaxis_rangeslider_visible=False)
    return fig

def color_boolean(val):
    if val is True:  return "background-color: lightgreen"
    if val is False: return "background-color: pink"
    return "background-color: lightblue"

print("Engine ready.")
Engine ready.
```

Analyze Data

The following pulls 300 daily candles and computes the following:

- Kalman Filter: a price smoothing noisy fluctuations, key indicator of close of day price strength
- Bollinger Bands: An elastic envelope derived from 14-day price average. Narrowing suggests quiet market, flaring suggest volatile.
- Ichimoku Spans: Two mean price projections used as trend map, suggesting uptrend or downtrends between 9 adn 26 days.

- AMAT: Archer Moving Average Trends used as binary indicators showing diverging fast and slow averages, or aligning showing sustained trend (1 = “trending up”).
- Relative Strength Index (RSI). Speedometer, above 70, price running fast risks overselling and then cooling off. Below 30 may mean due to bounce.
- Choppiness Index: High values suggest inertia, low suggest clean trend.

Key values in these metrics in sell and buy point variables above, copied here again for review:
`ema_crossover = ema_14 > ema_91`

The originals scanned every Binance.US ticker, which is slow and noisy. Here we take the most liquid spot /USDT pairs by quote volume. Raise TOP_N to widen the scan; each extra symbol adds roughly a second.

```
tickers = exchange.fetch_tickers()
pairs = [(s, d.get('quoteVolume') or 0) for s, d in tickers.items()
         if s.endswith('/USDT') and ':' not in s]
universe = [s for s, _ in sorted(pairs, key=lambda x: -x[1])[:TOP_N]]
today = datetime.now().strftime('%Y-%m-%d')
print(f"{today}: scanning {len(universe)} pairs")
universe

2026-06-18: scanning 20 pairs

['USDC/USDT',
 'BTC/USDT',
 'ETH/USDT',
 'WLD/USDT',
 'USD1/USDT',
 'SOL/USDT',
 'XRP/USDT',
 'ZEC/USDT',
 'BNB/USDT',
 'XLM/USDT',
 'NIGHT/USDT',
 'RE/USDT',
 'NEAR/USDT',
 'SPCXB/USDT',
 'FDUSD/USDT',
 'RLUSD/USDT',
 'UNI/USDT',
 'ASTER/USDT',
 'U/USDT',
 'XPL/USDT']
```

Rank Data

Run the engine over the universe into one table. The `try/except` skips symbols too new to have full indicator history. For example a token listed days ago has no Ichimoku cloud and skips these values instead of hiding them.

```
rows = []
for symbol in universe:
    try: rows.append(calculate_indicator(symbol)[1])
    except Exception as e: print(f"skip {symbol}: {type(e).__name__}")
results = pd.DataFrame(rows)
print(f"\nscored {len(results)} pairs | {int(results.Buy.sum())} buys | {int(results.Sell.sum())} sells")
results.head()

skip RE/USDT: KeyError
skip SPCXB/USDT: KeyError

scored 18 pairs | 4 buys | 6 sells
```

Symbol	Buy	Sell	Close	RSI	Ichimok9	Ichimok9	Ichimok26	EMA GT Kalman	Low GT Kalman
USDC/USDT	False	False	1.0009	57.78	1.0080	1.0080	1.0109	True	False
BTC/USDT	False	True	62958.0	34.91	66314.5	66314.5	70990.4	False	False
ETH/USDT	False	True	1711.1	38.62	1775.16	1775.10	1964.7	False	False
WLD/USDT	True	False	0.6431	66.55	0.5374	0.5374	0.4748	True	True
USD1/USDT	True	False	1.0009	63.39	1.0001	1.0001	1.0000	True	True

Symbol	Doji	Dragonfly	Gravestone	MACD	MACD Signal	MACD Buy	MACD Sell	MACD below zero
USDC/USDT	False	False	False	0.0000	-0.0000	True	False	False
BTC/USDT	False	False	False	-2437.6307	-2930.6	False	False	True
ETH/USDT	False	False	False	-83.5294	-106.67	False	False	True
WLD/USDT	False	False	False	0.0772	0.0632	False	False	False
USD1/USDT	True	False	False	0.0002	0.0001	False	False	False

```
top = (results.sort_values(['Buy','EMA_gt_Kalman','AMAT','RSI'],
ascending=[False,False,False,True]).reset_index(drop=True).head(10))
bottom =
(results.sort_values(['Sell','AMAT','RSI'],ascending=[False,False,False]).reset_index(drop=True).head(10))
print("ranked.")

if top['Buy'].any():
    print("Entry candidates showing bullish trends and Kalman lead prices:")
    for s in top.loc[top['Buy'], 'Symbol']: print(" ", s)
else: print("No long signals showing strongest-ranked names.")

top.style.map(color_boolean)

ranked.
Entry candidates showing bullish trends and Kalman lead prices:
  XPL/USDT
  USD1/USDT
  XLM/USDT
  WLD/USDT
```

	Symbol	Buy	Sell	Close	RSI	Chop	AMAT	Ichimoku_9	Ichimoku_26	EMA_gt_Kalman	Low_gt_Kalman	Doji	Dragonfly	Gravestone	MACD	MACD_Signal	MACD_Buy	MACD_Sell	MACD_below_zero
0	XPL/USDT	True	False	0.097900	56.580000	41.100000	True	0.091700	0.091700	True	True	False	False	False	0.002300	-0.001500	False	False	False
1	USD1/USDT	True	False	1.000900	63.390000	61.650000	True	1.000100	1.000000	True	True	True	False	False	0.000200	0.000100	False	False	False
2	XLM/USDT	True	False	0.234500	64.250000	47.100000	True	0.216300	0.218900	True	True	False	False	False	0.009500	0.007700	False	False	False
3	WLD/USDT	True	False	0.643100	66.550000	51.270000	True	0.537400	0.474800	True	True	False	False	False	0.077200	0.063200	False	False	False
4	USDC/USDT	False	False	1.000900	57.780000	17.340000	True	1.008000	1.010900	True	False	False	False	False	0.000000	-0.000000	True	False	False
5	NEAR/USDT	False	False	2.237000	52.850000	57.490000	False	2.348000	2.166000	True	False	False	False	False	0.053200	0.065300	False	False	False
6	SOL/USDT	False	False	69.710000	42.200000	50.440000	True	71.390000	79.270000	False	False	False	False	False	-2.784900	-3.853600	False	False	True
7	ASTER/USDT	False	False	0.638000	46.350000	39.450000	True	0.698500	0.695500	False	False	False	False	False	-0.004200	-0.008300	False	False	True
8	FDUSD/USDT	False	False	0.998400	48.610000	48.650000	True	0.998300	0.998000	False	False	False	False	False	-0.000000	-0.000100	False	False	True
9	U/USDT	False	False	1.000700	53.000000	66.910000	True	1.000900	1.000900	False	False	False	False	False	-0.000000	0.000000	False	False	True

A chart showing strongest buy candidate, with the Kalman and EMA-14 trend lines:

```
from IPython.display import HTML
HTML(plot(top['Symbol'].iloc[0]).to_html(include_plotlyjs="cdn", full_html=False))
<IPython.core.display.HTML object>
# Save the chart as a standalone, self-contained HTML file you can open in any browser.
fig = plot(top['Symbol'].iloc[0])
fig.write_html('outputs/chart.html', include_plotlyjs=True, auto_open=False)
print('saved outputs/chart.html')
saved outputs/chart.html
```

Performance Metrics

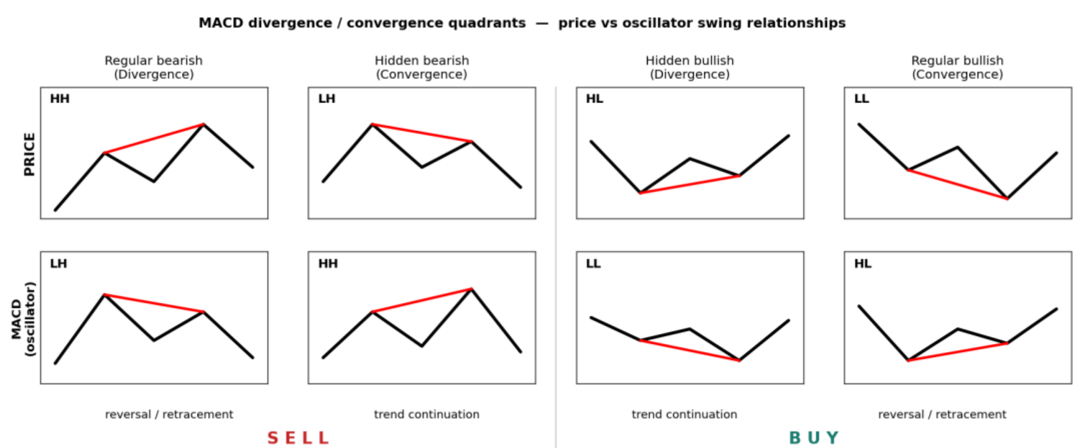
MACD Trends

stands for Moving Average Convergence Divergence. Gerald Appel, 1970s. It is a momentum and trend tool, plotted in its own panel below price rather than on the candles. Because the MACD line is fast EMA minus slow EMA, it crosses its own zero line at the same moment the 12-EMA crosses the 26-EMA on the price chart. These `crossovers` are key indicators in day trading, often sound buy points where MACD crosses above or sell points where crosses below. Additionally, the zero-line cross occur during bullish context, with below zero suggesting change to bearish trend. However, the signal-line tends to occur earlier than the zero line cross that offers greater notice of opportunity. Trade off is falling into more false starts.

Beneath the line graph, the histogram can add to our predictions providing distances between lines that guide us to whenm lines cross. The height of these distances represents momentum, so that a growing histogram indicates accelerating divergence. Alternatively, a shrinking histogram of converging lines suggests momentum is fading signalling a likely turn.

Divergence presents important metric in day trading that is least mechanical to read. Observing changing swings between MACD highs and lows, we can identify a shift in trends is expected. Bearish divergence can be characterised by higher highs while MACD show lower highs that caution an uptrend is hollow. aternatively, a bullish divergence can appear with lower lows while MACD presents higher lows that caution a downtrend is exhausting. Need guidance on this, but seems
divergence is mostly used to protect a position before change, rather than a precise entry to trade.

Diverge/Convergence Quadrants



The full taxonomy compares the last two price swings against the matching MACD swings. Notation: HH higher high, HL higher low, LH lower high, LL lower low.

Price	MACD	Source term	Standard term	Reading	Action
Higher high	Lower high	Divergence	Regular bearish	reversal / retracement	SELL
Lower high	Higher high	Convergence	Hidden bearish	trend continuation	SELL
Higher low	Lower low	Divergence	Hidden bullish	trend continuation (up)	BUY
Lower low	Higher low	Convergence	Regular bullish	reversal / retracement	BUY

Two cautions from the research. First, terminology is not standardised: GoodCrypto defines "divergence" as price stronger than the indicator and "convergence" as price weaker, which is the geometry above but the opposite word to how many texts use them. Read the HH/HL/LH/LL pattern, not the label. Second, a single chart can show mixed signals: in their BTC example the peaks diverged while the troughs converged, netting out bearish only because the peak structure dominated. Weigh both swing series, do not act on one line in isolation.

MACD in Crypto Markets

MACD is well respected in BTC and ETH markets, but cannot be handled in isolation. It is recommended to consider alonside position sizing and a expected stop. Some suggest a good crypto technique should draw the trendline on the MACD itself, not only on price. For example, in the May 2021 BTC top, price held a rising trendline while the MACD was already tracing a falling one. The MACD trendline broke first. The same setup appeared on the SOL 4h chart, pushing higher highs into resistance while MACD showed clearly lower highs. Was this a textbook bearish divergence, as it preceded the stall?

The choice of timeframe is important, but does not change whether MACD works; a 4h or 1h MACD simply updates faster than a daily one. The real limiter is volatility: the more violent the asset, the less any indicator forecasts cleanly, so widen confirmation in fast markets.

Corroborating Metrics

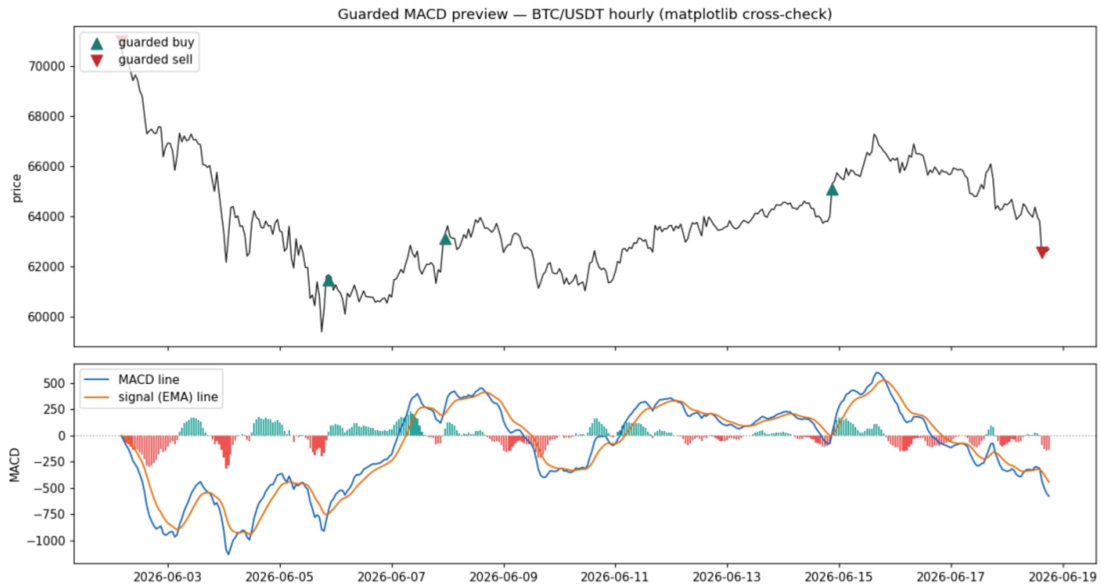
MACD gives momentum but poor at signaling overbought/oversold thresholds, so traders may confirm crossover against a second indicator and only act when both agree. In rough order of their selective applications:

Partner	What it adds	Entry rule	Exit
Relative Vigor Index	closing strength vs range	both cross same direction	MACD opposite cross
Money Flow Index	price + volume, few signals	MFI overbought/oversold then MACD cross	MACD opposite cross
TEMA (50)	triple-smoothed trend	price breaks TEMA and MACD crosses	contrary signal from both
TRIX	momentum oscillator	MACD cross matched by TRIX zero-cross	MACD cross, or looser, TRIX zero-cross
Awesome Oscillator	5/34 SMA momentum	MACD cross confirmed by AO	both turn contrary
MA (20)	trend validation	price tests the 20-MA, then MACD crosses up	—

The shared idea is the guardrail we already build into the metric: do not take a bare crossover, require corroboration. A second indicator is one form of that; our epsilon noise band and confirmation bars are another, internal to MACD itself. MFI's volume gate is the most distinct addition and the strongest candidate to add next, since the histogram and slope already cover much of what RVI, TRIX, and AO contribute.

Final Metrics

The current iteration targets these metrics by defining the following variables, including `cross_up` and `cross_down` crossover points; as well as `guarded_buy` and `guarded_sell` constraints. These were applied as conservative measures around noise which we found recommended in majority of readings. Histogram variables were derived for both `hist_slope` and `converging` flags to highlight early fade nearing zero. Divergence points were constructed at swing pivot points for variables of `bear_div` and `bull_div` representing bearish and bullish quadrant data of regular bounds. MACD breaks and hidden divvergence, which are not yet coded, are next target. `strength` is compiled from band clearances, slope spikes, zero field, and corroboration by divergence, which was numerically viewed as significantly weighted swing series.



BTC preview

Entry Points

Buy trends are displayed first with Kalman flags displayed and lowest RSI scores at the top. Sell points are presented as the mirror to these, which is also characteristic of how the computational checklist was built between them.

Exit Points

```
if bottom['Sell'].any():
    print("Exit candidates trending down showing prices below Kalman:")
    for s in bottom.loc[bottom['Sell'], 'Symbol']: print(" ", s)
else: print("No short signals today; showing weakest-ranked names.")
bottom.style.map(color_boolean)

Exit candidates trending down showing prices below Kalman:
  ZEC/USDT
  NIGHT/USDT
  XRP/USDT
  ETH/USDT
  BNB/USDT
  BTC/USDT
```

	Symbol	Buy	Sell	Close	RSI	Chop	AMAT	Ichimoku_9	Ichimoku_26	EMA_gt_Kalman	Low_gt_Kalman	Doji	Dragonfly	Gravestone	MACD	MACD_Signal	MACD_Buy	MACD_Sell	MACD_below_zero
0	ZEC/USDT	False	True	455.840000	46.690000	43.040000	False	470.817500	470.060000	False	False	False	False	False	-14.988900	-18.813400	False	False	True
1	NIGHT/USDT	False	True	0.031100	44.760000	61.700000	False	0.033800	0.035200	False	False	False	False	False	-0.000700	-0.000500	False	False	True
2	XRP/USDT	False	True	1.146500	40.390000	48.660000	False	1.200700	1.299900	False	False	False	False	False	-0.038100	-0.048400	False	False	True
3	ETH/USDT	False	True	1711.110000	38.620000	51.150000	False	1775.160000	1964.710000	False	False	False	False	False	-83.529400	-106.668800	False	False	True
4	BNB/USDT	False	True	578.480000	37.600000	54.720000	False	627.025000	651.100000	False	False	False	False	False	-12.618900	-11.667000	False	True	True
5	BTC/USDT	False	True	62958.010000	34.910000	53.050000	False	66314.515000	70990.455000	False	False	False	False	False	-2437.630700	-2930.637400	False	False	True
6	WLD/USDT	True	False	0.643100	66.550000	51.270000	True	0.537400	0.474800	True	True	False	False	False	0.077200	0.063200	False	False	False
7	XLM/USDT	True	False	0.234500	64.250000	47.100000	True	0.216300	0.218900	True	True	False	False	False	0.009500	0.007700	False	False	False
8	USD1/USDT	True	False	1.000900	63.390000	61.650000	True	1.000100	1.000000	True	True	True	False	False	0.000200	0.000100	False	False	False
9	UNI/USDT	False	False	3.209000	60.980000	32.590000	True	3.035000	3.243000	False	True	True	False	False	-0.033200	-0.142900	False	False	True

Four Votes

MACD is derived as guarded crossover points in the `macd_lab` metrics which is presented in terms of a regime going forward. That is, after a guarded buy the stance stays +1 until a guarded sell flips it. MA crossover is the plain fast-over-slow test from the MA-crossover notebook, +1 when the 20-period SMA is above the 50, -1 below.

Fibonacci is contextual: on a rising leg, where price pulled back into the 0.5-0.618 golden pocket votes +1 to buy the dip; on a falling leg, price bounced into that pocket votes -1 (sell the rip); otherwise 0. Candles are an adjusted but assuming pattern, where +1 engulfs on a bullish and -1 on bearish, held live for a few bars before lapsing.

Worth noting that we detected a real bug in Candle module, a positional column rename that swapped open and close, so its pattern was informed by invalid fields. The version here uses the standard engulfing definition derived from valid columns.

2-3-Rule Thresholds

We aimed for two-point agreement as the standard rule, and used the stricter three-point version when a setup had fewer trades to learn from. Localised bandwidths were tuned tightly to conditions, and the backtest only ever used information that would have been available at the time, nothing from hindsight. A signal fires the moment the score crosses the threshold, so the dashboard shows more markers than the backtest actually trades. That gap is expected: if a second buy signal fires while we already hold a position, it doesn't open a new one.

```
score = w_macd*MACD + w_ma*MA + w_fib*Fib + w_candle*Candle      (weights default 1)
BUY  fires when score first reaches +threshold   (default +2)
SELL fires when score first reaches -threshold   (default -2)
```

Honest Cynicis Check

Hourly data, the ten-coin universe, about 1000 bars (~41 days), 0.1% fee per side, long-flat, threshold 2. This window was a broad crypto downtrend.

coin	strategy	buy & hold	trades	win %	max DD
BTC	-1.4%	-21.3%	4	50	-7.3%
ETH	-9.2%	-26.1%	5	20	-16.1%
SOL	-19.8%	-22.1%	6	33	-36.9%
BNB	-0.9%	-9.9%	5	40	-6.7%
XRP	-6.3%	-17.9%	4	50	-15.5%
ADA	-16.9%	-38.2%	5	20	-22.0%
AVAX	-8.0%	-33.6%	6	33	-16.7%
LINK	-19.7%	-20.3%	7	29	-25.4%
LTC	-10.1%	-23.4%	4	25	-17.1%
DOGE	-9.2%	-22.6%	3	33	-15.1%
mean	-10.1%	-23.5%			

Read it plainly. The engine lost money on every coin, but lost roughly half of what holding lost, because staying flat through the downtrend avoided the worst of it. That is a drawdown-reduction property, not an edge. Beating buy-and-hold by being absent during a fall is easy and does not survive into a rising or sideways market, where sitting flat means missing gains. Win rates of 20 to 50 percent confirm there is no demonstrated predictive skill here yet. This matches the project's standing NO-GO finding: cleaner instrumentation, still no proven edge.

These are in-sample numbers over a single short, one-directional window. They tell you the rule does something coherent, not that it will keep working.

Change of Heart

Walk-forward is the next step the data has now earned: split each coin's history into rolling train and test segments, choose the weights and threshold on train only, score once on the untouched test segment, repeat forward, and report only the out-of-sample aggregate with fees. Test across regimes, not just this falling one, so a bull and a sideways stretch are included. Add a stop and a take-profit, since a flat-long rule with no risk control flatters drawdown. Only if the out-of-sample, multi-regime, after-fees result clearly beats both buy-and-hold and a coin-flip is any of this worth real money, and even then the operator owns the decision and the live switch stays off.

```
stamp = datetime.now().strftime('%Y%m%d')
path = f'./outputs/DailySignals_{stamp}.csv'
results.to_csv(path, index=False)
print("saved", path)
saved ./outputs/DailySignals_20260618.csv
```

Decision Checklist

1. Read balances (USDT, BTC, ...).
2. Size each trade as cash divided by the number of long candidates.
3. Act on exits (`bottom`) before entries (`top`).
4. Widen timelines and trend comparisons (all data saved & dated in /outputs/).

Adjust Report Formats (bash terminal)

- `quarto preview day-metrics.ipynb --no-execute`
- `quarto render day-metrics.ipynb --to html --no-execute`
- `quarto render day-metrics.ipynb --to docx --no-execute`
- `quarto render day-metrics.ipynb --to pdf --no-execute`